Fiscal Year:	FY 2017	**Task Last Updated:**	FY 08/16/2016
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Project Title:	Collection of Immune/Stress-related Tissues from Mice Flown on ISS		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2014 Space Biology Flight NNH14ZTT001N
Start Date:	11/01/2014	**End Date:**	11/30/2018
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA ARC
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Flight Program:	ISS		
Flight Assignment:	Tissue Sharing NOTE: End date changed to 11/30/2018 per F. Hernandez/ARC (Ed., 2/7/17) NOTE: NSSC shows end date of 11/30/2017 (Ed., 8/3/16) NOTE: Extended to 10/31/2016 per A. Chu/ARC and NSSC information (Ed., 10/28/15)		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):	Gridley, Daila Ph.D. (Loma Linda University) Pecaut, Michael Ph.D. (Loma Linda University)		
Grant/Contract No.:	NNX15AB45G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NNX15AB45G was funded in 2014 in response to section B.7 - Special topic - ISS Rodent Tissue Sharing Opportunity announced as part of NRA NNH14ZTT001N. NASA indicated that a total of 20 mice were planned to be flown to the International Space Station (ISS) and 40 mice would serve as ground controls. Four immunization treatment groups were part of the experimental protocol that was designed to assess the in vivo antibody responses during spaceflight. In particular, the antibody response and the B cell immunoglobulin heavy-chain (IgH) repertoire would be determined in the splenic B cells from each of the treatment groups and compared among flight and ground treatments. Our group (Pecaut and Chapes) proposed to collect mouse liver, bone marrow, kidney/adrenal glands, and thymus to accomplish two specific aims. 1. Specific Aim 1: Collect thymus to weigh the tissue to compare relative thymic atrophy among treatment groups and adrenal glands to assess catecholamines. These tissues would have been used to assess short- and long-term stress to the flight animals. 2. Specific Aim 2: Collect liver and bone marrow. These tissues would be used for the preparation of RNA to allow for a comparative analysis of B cell IgH usage between lymphoid compartments. With the loss of astronaut work time assigned to Space Biology and > or = 2 year delay in the transport of mice to the International Space Station, we were approved to modify the aims of NNX15AB45G as follows: 1. Obtain liver tissue/RNA from the mice flown on Rodent Research 1 (RR1) verification flight. Preflight verification tests done as part of NNX15AB45G have shown that usable RNA cannot be obtained from bone marrow from frozen carcasses (using the freezing procedure currently used on ISS). Therefore, we proposed to focus on liver tissue. Liver RNA was available from the liver tissue (10 mice; 5 flight and 5 AEM ground controls). We proposed to do MiSeq next generation sequencing on a group of flight and ground samples. In addition, NASA Gene Lab had done HiSeq next generation sequencing on some of the liver tissue flown on RR1. We proposed to obtain those data. We proposed to determine IgH usage of the liver B cells in the mice (flight vs. ground controls). One of our goals is to determine if there is tissue specificity in B cell IgH usage. 2. An antiorthostatic suspension (AOS) was proposed to skeletally unload mice to simulate the skeletal unloading that occurs during spaceflight (This experiment will be done as part of NNX13AN34G). In this protocol, mice will be subjected to AOS. Two weeks later the mice will be immunized in 1 of 4 ways; PBS, CpG only, tetanus toxoid (TT) only, and TT with CpG. Injections will be intraperitoneal (i.p.). Two weeks after immunization and continued AOS, we will obtain liver tissue from those mice. We will directly assess the changes in IgH usage caused by immunization ± adjuvant (CpG) and will address the impact of skeletal unloading. One of our original goals was to examine the impact of spaceflight on host B cell responsiveness in different tissue compartments. Therefore, this experiment was expected to partially fulfill the goals of Aim 1 of the original grant by using AOS as a spaceflight model. We proposed to compare the impact of skeletal unloading using AOS on host responsiveness to antigen challenge and address the issue of tissue specificity in B cell IgH usage. In addition to the IgH usage studies, we proposed to collect thymus to weigh the tissue to compare relative thymic atrophy among treatment groups and adrenal glands to assess catecholamines. The data obtained from these tissue analyses will be used to assess short- and long-term stress to AOS animals.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Spaceflight offers a unique environment that impacts many physiological systems. This occurs not only in humans but also in rats, mice, and rhesus monkeys. Among these systems, bone and the immune response are connected and are known to be significantly altered by the skeletal unloading of spaceflight. B lymphocytes are one of the cells that develop in the bone marrow. They make antibodies in response to infection or immunization with vaccines. Moreover, the B cells that protect people from disease after receiving immunizations localize to the bone marrow as well as other lymphoid organs. We will obtain valuable information about what changes occur in the B-cell populations that develop in lymphoid organs such as liver and compare the development of these B-cell populations found in other lymphoid compartments. This understanding will tell us if more precautions are necessary on Earth when people are skeletally unloaded. On Earth, skeletal unloading occurs to individuals that are bedridden for long periods of time, wheel-chair bound, or lose the ability to stand for long periods of time. Therefore, the space environment provides an accelerated view of what will happen to individuals that must endure these prolonged periods of skeletal unloading. This experiment will have direct relevance to individuals on Earth.
	Assessment of the Light Chain Antibody Repertoire in RR1 Validation Flight Liver Tissue B cells can facilitate host response to pathogens through cell surface expression and secretion of antibodies. In both mice and humans, B cell development first occurs in the fetal liver and later migrates to the fetal bone marrow where B cell development will continue throughout adulthood. While many studies have assessed the antibody repertoire within the fetal liver, little is known about the B cell population in an adult liver. In order to assess the antibody repertoire, our lab examined the composition of gene segments that are joined together to form a heterodimer of heavy and light chains that make up an antibody. Heavy chains are comprised of one Variable (V), one Diversity (D), one Joining (J) gene segment in addition to a constant region, while light chains lack the diversity region. We characterized the antibody repertoire of adult mouse liver through the use of data and samples collected from the RR1 validation flight. Rodent Research 1 included two separate mouse studies: NASA (N=7 ground; N=7 flight, 37 days in orbit) and CASIS (N=3 ground; N=3 flight, 21-22 days in orbit). Illumina HiSeq data (50 bp, 1x) from liver RNA was provided by the NASA GeneLab. Additionally, Illumina MiSeq sequencing was performed on RNA isolated from the liver and spleen of CASIS mice at the Kansas State University Integrated Genomics Facility for Illumina MiSeq sequencing (2x, 275-800bp). Illumina sequencing data was imported into CLC Genomics Workbench and quality cleaned. Reads were then mapped to reference variable gene segment sequences obtained from ImMunoGenetics (IMGT) and the National Center for Biotechnology Information gene database. Fasta files for reads mapping to reference sequences were then submitted to the IMGT HighV-Quest tool for junctional analysis. Sequences obtained from HiSeq were not compatible with HighV-Quest due to the short sequence length and as a result only the variable gene segment composition can be assessed within these datasets. HighV-Quest output files from MiSeq sequences were cleaned to remove duplicates resulting from mapping bi-directional sequencing reads and gene segment composition was assessed. The composition of V gene segment usage obtained from HiSeq reads was assessed by ground and flight conditions for both ground and flight conditions for both NASA and CASIS cohorts. Differences in gene segment usage were assessed using the Mann-Whitney U-test. Briefly, both cohorts exhibited a change in light chain variable gene segment usage,

Task Progress:	while only the CASIS cohort shown a difference between heavy chain V gene segment usage (abstract submitted for the 2016 ASGSR annual meeting). At present, the MiSeq data is available for CASIS flight animals and CASIS ground samples are currently being sequenced. We have discovered that a significant challenge of assessing the antibody repertoire of liver is the low abundance of sequenced reads mapping to variable gene segments. For example, 1.9% and 2.5% of CASIS flight Illumina MiSeq sequencing reads obtained from the spleen mapped to heavy and light chain references respectively, while both HiSeq and MiSeq reads mapped 0.005-0.1% to reference sequences. Upon IMGT submission and removal of duplicate reads in MiSeq data, just under 200 Kappa light chain V gene segments and less than 30 Lambda chain V gene segments could be identified. The impact of spaceflight on the repertoire of antibody light chains can be assessed upon the completion of ground control MiSeq sequencing, adding to HiSeeq data. In addition to the comparison of ground control and flight spleen samples, we will also be able to assess whether the antibody populations vary by tissue using RNA isolated from spleen tissue of the same animals; however, findings may be inconclusive due to the limited population of light chain gene segments identified in the liver. Previously described assessments utilize unimmunized mice; however, our lab will have the opportunity to assess the impact of spaceflight-like conditions in immunized mice using a ground based suspension model. We anticipate completing RR1 comparisons this fall. Assessment of the Heavy Chain Antibody Repertoire in RR1 Validation Flight Liver Tissue NASA Mice: The most common V gene segment used among all three groups is V3-01 totally over 50% of the flight repertoire and 40% of the ground repertoire. V1-16 was the next most common with just under 10% of the repertoire for both groups followed by V1-56, which comprised around 5% of the repertoire for both groups. The most common constant region was IgG2b totaling almost 40% of the flight mouse and over 50% of the ground mouse repertoire. IgE comprised 17% and 12% of the repertoire of flight and ground mice respectively, with IgG3 as the next most common at 13% and 11% of the flight and ground animals, respectively. CASIS Hi-Seq: The most common V gene segment used was V3-1, with an overall usage around 20% for all groups. The next most common was V1-16 with 11% of the repertoire for the basal group and around 8% for both the flight and ground groups. V1-56 the third most common V gene segment comprising between 7-8% of all groups. IgG2b was the most common constant region with over 40% of the repertoire for the ground control, 50% for the basal control, and almost 60% for the flight group. IgM was the next most highest with 20% of the ground control group, 15% for the basal group, and less than 10 for the flight group. All other isotypes were below 10% for all groups. Casis MiSeq: The resequencing of the CASIS animals has yielded very low liver results. While spleens from these same animals using the same techniques are reaching over 20,000 sequences, flight animal 1 had 40 results, flight two had 48, and flight 3 had only 20 productive sequences. One hypothesis for the low numbers of reads could be because the adult liver does not have many B cells in situ residence or trafficking through in the vasculature. We have completed sequencing the ground control animals but these have yet to be analyzed. For flight animals, the most common V gene segment used was V3-6 with 18%, 29%, and 19% of the total repertoire in flight 1, flight 2, and flight 3 respective. V13-2 was the next most common with 8% for flight 1 and 3% for flights 2 and 3. V1-69 was the next most common with 4%, 2%, and 7% of the repertoire. D1-1 was the most common D gene used at around 20% for all animals, however an average of 40% of the D gene segments used were unable to be identified. The J gene segments are fairly balanced in usage among all animals with J2 and J4 being the most common. Due to low read numbers, constant region data are relatively unreliable with as few as two constant regions identified for some animals. IgM was the most common with 65% to 100% of constant regions identified. The average CDR3 length is between 11 and 12 amino acids and no CDR3 overlap was detected among the three groups. We anticipate completing the ground analyses in the next couple of months to complete comparisons between flight and ground mice and assessing the differences/similarities among MiSeq and HiSeq data. Antiorthostatic Suspension Study The AOS was done to skeletally unload mice to simulate the skeletal unloading and stress that occurs during space flight (Experiment 1 took place in March 2016 [5 mice per treatment group]; Experiment 2 took place in May 2016 [5 mice per treatment group]). In this protocol, female, 8-week-old mice were subjected to AOS. Two weeks later the mice were immunized in 1 of 4 ways; PBS (buffer control), CpG only (adjuvant), tetanus toxoid (TT; antigen) only, and TT with CpG. Injections were done by intraperitoneal (i.p.) route. Two weeks after immunization and continued AOS (28 day total unloading), we obtained both spleen and liver tissue from those mice. This protocol was followed so that when mice are flown to ISS in the future, we will have the AOS as a comparison group with parallel variables. Liver samples were flash frozen in liquid nitrogen and RNA was purified using Trizol and isopropanol precipitation methods. We have obtained 80 liver samples from these mice and over 80% of the RNA samples have RINS = 8.0. Sequencing and data analyses will proceed during the next project period. We will assess the changes in IgV (heavy and light chain) usage caused by immunization ± adjuvant (CpG) and will address the impact of skeletal unloading. Impact of immunization and antiorthostatic suspension (AOS) on stress markers. Preliminary analysis of the stress marker, corticosterone, suggest that while there was an impact of unloading on circulating corticosterone, there was no impact of TT or CpG. We also assessed thymus mass on the AOS mice to assess long-term stress on the mice. We do see a significant decrease in mass due to HLU, which is consistent with the corticosterone response.
Bibliography Type:	Description: (Last Updated: 09/15/2021)
Abstracts for Journals and Proceedings	Rettig T, Bye B, Pecaut MJ, Chapes SK. "Characterization of the normal mouse IGH antibody repertoire." Presented at the 2016 American Association of Immunologists Meeting, Seattle, WA, May 13-17, 2016. 2016 American Association of Immunologists Meeting, Seattle, WA, May 13-17, 2016. Program #64.7. https://immunology2016.zerista.com/event/member/257567 ; accessed 8/16/2016. , May-2016

Abstracts for Journals and Proceedings	Hlavacek S, Chapes SK. "Studying the overexpression of PRDM16 in mouse cells." Presented at the 6th Biennial National IDeA Symposium of Biomedical Research Excellence (NISBRE), Washington D.C., June 26-28, 2016. 6th Biennial National IDeA Symposium of Biomedical Research Excellence (NISBRE), Washington D.C., June 26-28, 2016. https://www.eiseverywhere.com/ehome/nisbre/387913/ to search for abstract. , Jun-2016
Abstracts for Journals and Proceedings	Nishiyama NC, Rettig T, Ward C, Gridley DS, Chapes SK, Pecaut MJ. "Simulated Microgravity on Primary Antibody Response." To be presented at the 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016. 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016. https://www.asgsr.org/index.php/2016-annual-meeting , Oct-2016
Abstracts for Journals and Proceedings	Ward C, Rettig T, Pecaut MJ, Chapes SK. "Impact of Spaceflight on the Antibody Repertoire of Unimmunized Mice." To be presented at the 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016. 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016. https://www.asgsr.org/index.php/2016-annual-meeting , Oct-2016